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To cite this article: P. Lavado, S. Cueto, G. Yamada & M. Wensjoe (2019) The Effect of a Charter School in Peru (Fe Y Alegría) on School Achievement: Exploiting a School Lottery Selection as a Natural Experiment, *Studies in Economics and Econometrics*, 43:3, 95-110, DOI: [10.1080/10800379.2019.12097352](https://doi.org/10.1080/10800379.2019.12097352)

To link to this article: <https://doi.org/10.1080/10800379.2019.12097352>



Published online: 09 Nov 2020.



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THE EFFECT OF A CHARTER SCHOOL IN PERU (FE Y ALEGRÍA) ON SCHOOL ACHIEVEMENT: EXPLOITING A SCHOOL LOTTERY SELECTION AS A NATURAL EXPERIMENT¹

P. Lavado^{2*}, S. Cueto³, G. Yamada² & M. Wensjoe³

Abstract

This study estimates the effect of one charter school (a public-private partnership) on mathematics and reading comprehension among second grade students in Peru between 2007 and 2012. The study uses an identification strategy to estimate the causal effect of a charter school. The strategy is based on a natural experiment of an admission lottery to determine which students would be accepted into second grade at the charter school. The results show that the charter school achieved substantial gains in the scores of the lottery winners that are equivalent to 0.4 standard deviations. We also find that this effect has increased over time.

1 Introduction

Public-private partnerships (PPP) are taking an increasingly key role in improving the quality of education. However, evidence on its impact is scarce (Patrinos, Barrera-Orsorio & Guáqueta, 2009). PPPs are particularly important in developing countries because government inefficiencies often reduce the quality of education (Angrist, Bettinger, Bloom, King, & Kremer, 2002).

¹ In memoriam Jesús Herrero SJ. This study was possible thanks to the support of the Inter-American Development Bank. The authors are grateful for the generous cooperation of Fe y Alegría Perú, and in particular its Director Javier Quiróz. We also thank the Educational Quality Measurement Unit at the Ministry of Education and its Director, Liliana Miranda, in particular. We also are grateful for the valuable comments by Julián Cristia, an anonymous IDB reviewer, two anonymous reviewers and from the participants at the Peru's Economists Association Conference. César Salinas, Ana Paula Franco, and Jesús Gutiérrez gave us valuable support in the statistical analysis, while Varinia Arévalo assisted in data gathering.

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PPPs were born from the collaboration of states with religion organizations, private providers, or non-governmental organizations (NGOs). These partnerships can take different forms. Financing is shared between the government and the private organization in which governments finance teacher salaries, while private organizations finance, for example, infrastructure and management (Lewis & Patrinos, 2012). This scheme ensures competition in the education sector because of the increase in the teacher's training, better infrastructure and school materials, and better-equipped classrooms (LaRocque & Patrinos, 2007). The main characteristic is that school management gains some degree of autonomy, with more flexible rules than in public schools. This study focuses on a charter school, which is a partnership in which the government provides financing while the private sector delivers educational services to students.

The remainder of this paper is organized as follows: The following section presents a literature review. Section 3 presents the identification strategy, and section 4 presents the data, empirical strategy used for the analysis, and the results. Then, we private empirical results and the comparison between Fe y Alegría and public or private schools in section 5. Section 6 concludes.

2 Literature review

Studies show a positive effect of PPPs on learning outcomes. In the United States, Hoxby and Rockoff (2004) find 10 to 11 percentage point improvements in the early grades, for charter schools in Chicago compared to public schools. By contrast, in New York, Hoxby, Murarka and Kang (2007) find that the effect of a charter school is evident for the lowest grades in math and reading comprehension (0.09 and 0.04). This effect is maintained in the long run for schools in Arizona, Florida, and Texas (Booker, Gilpatric, Gronberg & Jansen, 2007, Bettinger, 2007, Hanushek, Kain, Rivkin, & Branch, 2007). In a case in Latin America, based on the Concession Schools program, Barrera-Osorio (2006) conclude that in private schools that provide public education in Colombia, there are positive effects on math and reading comprehension (0.19 and 0.27 respectively).

In this context, the main objective of this study is to estimate the impact of a worldwide PPP (Fe y Alegría, FyA) on children's learning. Despite the effort devoted by the FyA to enhance children's education from low income households, there is no evidence of a rigorous assessment of this experience. FyA is selected because it is the largest PPP for education in the world.

FyA uses an autonomous school administration with public financing and manages schools in Spain, Latin America and Africa. Founded in 1960, it currently operates 1,240 schools that serve 1,473,074 children in 20 countries (Fe y Alegría, s.f). Its target population is children from low income households.

The primary attribute of FyA is the autonomy that it grants to schools in choosing their administration (including the school director) and teaching staff. Teacher and administrator salaries are paid by the state, which also pays for teaching materials and school facilities (Alcázar & Cieza, 2002). FyA partly pays maintenance and other current expenses through a number of activities, initiatives, and donations, which are mainly from Europe. FyA manages its human resources by recruiting, training, and evaluating them based on their performance and the children's learning.

Another attribute is FyA's close relationship with parents. In particular, parents must attend academic follow-up meetings, classroom committees, and school general meetings. Also, they help in fund raising through a number of social events. In addition, three Sundays every year they must help to improve the facilities (e.g., fixing classroom ceilings, painting walls, and repairing chairs and tables). Parents must also make two payments to the school. The first, worth 50 Peruvian soles (approximately 14.64 USD²), is their annual contribution to the parent association to comply with the regulation these associations must follow to be involved in public educational institutions (Law 28628). The second payment is made by parents of primary school students of 40 soles (11.71 USD) annually that covers teaching materials (photocopies) and finances activities (school sports competitions, Mother's Day, etc.). Families who cannot afford those payments are exempted, and school pays their contributions. These payments are not required for children's enrollment.

The main hypothesis of this study is that the charter school achieves educational gains in math and reading comprehension compared to comparable public or private schools. Our second main hypothesis states that those gains are higher when compared to private schools. This hypothesis is supported by the fact that Peru has experienced an exponential growth of low-cost private schools of low quality. These schools have been attracting prospective students from public schools based on an illusion that private education is better than public education. In general, better management, equipment, infrastructure, and public services in the charter school are possible mechanisms that could explain the results.

² The current exchange rate is 3.42 soles per dollar.

3 Identification strategy

The main identification strategy for estimating the causal effect of the charter school is a natural experiment based on an admission lottery done in a typical FyA school in Lima (Peru). FyA schools are in high demand because of their good reputation and tuition-free education. Consequently, because of the excess of applicants, they follow a sequential selection. First, potentially interested parents are invited to register. Alumni's children or students' siblings are admitted directly. The next priority goes to district residents, particularly indigent applicants. Finally, when the number of applicants exceeds the available vacancies, students are admitted by lottery. The lottery takes place at the school in an open meeting for all registered parents. The applications are extracted one by one from a box until completing the number of available openings. Another 10 or 15 additional numbers are also extracted. The students whose numbers are selected in the lottery are offered a place. If parents do not deliver the required documents in time or fail to register for some reason, the place is offered to the first additional number chosen. We used lotteries from 2006 to 2011. These random lotteries allow for the estimation of the impact that FyA has on school performance in math and reading comprehension from a Census evaluation implemented by the Ministry of Education. This process addresses the issue of selection bias that occurs in other studies (Parra & Wodon, 2009).

The main results show a positive and statistically significant effect of the FyA of 0.39 and 0.41 standard deviations (s.d.) on math and reading comprehension, respectively. When compared with public or private schools, the effect is -0.37 s.d. on the private school and is -0.5 sd on the public school for reading comprehension. While for math, the effect is -0.36 s.d. on the private school and is -0.5 s.d. on the public school. The effects are similar or even larger than those found in previous studies. Angrist, Dinarsky, Kane, Pathak and Walters (2010) estimate effects of 0.2 and 0.4 standard deviations in English and math, respectively, for charter schools in Massachusetts. Lottery studies of charter schools in Harlem (Dobbie & Fryer, 2011) have found an effect of 0.2 s.d. in math and English, while Curto and Fryer (2011) find similarly large effects in a Washington, DC charter school (0.2 s.d. in reading and 0.23 s.d. in math).

4 Data and empirical strategy

The school under study was founded in 1980 in one of Lima's most densely populated and poorest districts. The district has 12% of the capital city's population. The local poverty rate is estimated at 24% compared to Lima's 5%. All the students live in that district. The facilities of the FyA school under study comprise 24 primary grade classrooms and 25 secondary school classrooms with an average of 35 students per class. The school's management includes a superintendent, a primary deputy superintendent, the secondary school coordinator, and the pastoral coordinator that are all women. The school operates in two shifts. The morning shift runs from 7:30 to 13:05 and accommodates the high school classes, while the primary school students attend from 13:15 to 18:05. A total of 66 teachers teach at the school: 28 in primary school and 38 in the high school. Primary classes are taught by one (male or female) teacher each who are charged with teaching all each class's courses. In high school, although each class has an appointed tutor, classes are taught by several teachers (each teaching a specific subject). The school runs four workshops for high school students: dressmaking, woodworking, office automation, and electricity repairs. All students must attend the four workshops in their first two years of high school and then choose the area where they will specialize in their last three years.

We started by reviewing the list of applicants to the FyA school for the years 2006, 2007, 2008, 2009, and 2011. There are 543 students registered for those lotteries. The list of winning students was sent to the Ministry of Education's Quality Measurement Unit. The Ministry returned an anonymous listing with the results of the math and reading comprehension tests of second grade students (the Students Census Evaluation). The Ministry of Education's listing covered 462 students, 328 were admitted at the FyA school under review, 85 were admitted to other public schools, and 49 enrolled in private schools³. This is an interesting result because it shows that a number of families decided to pay tuition at a private school rather than send their children to a public school when unable to register at the FyA. There was a total of six students for all years who declined to attend the FyA school even though they had won the lottery, approximately 2% of winning students.

The FyA lottery may be regarded as a social experiment. In this scheme and following the evaluation parameters (power of 80%, significance level at 95%, 70% treated, and 462 total observations), the minimum detectable effect is set at 0.25 s.d. Using the same calculation but only comparing FyA with other public schools, the

³ See Appendix N° 1.

minimum detectable effect is 0.30 s.d. The same comparative calculation between FyA and other private schools shows a minimum detectable effect of 0.37s.d.

Table 1 has a set of descriptive statistics gathered before the lottery between winning students and non-winning students for years 2013, 2014, and 2015. Similar records are not available for other years because either data had not been gathered or records were lost. In turn, census evaluation results are not available for children tallied in the descriptive statistics.

For lottery participants, the variables before treatment should be balanced. In fact, there were no ex ante statistically significant differences for most socioeconomic variables between students eligible to attend FyA and students who were not eligible. Significant differences were found in the mother's education.

Table 1: Descriptive statistics

VARIABLES	(1) Admitted	(2) Not admitted	(3) Admitted versus not admitted ^a
Male	0,473 (0,500)	0,484 (0,501)	-0,012 (0,050)
Father lives at home	0,906 (0,292)	0,852 (0,356)	0,054* (0,031)
Mother lives at home	0,978 (0,146)	0,963 (0,189)	0,015 (0,016)
Single household family	0,426 (0,495)	0,463 (0,500)	-0,037 (0,049)
Cement and mortar house	0,856 (0,352)	0,833 (0,374)	0,022 (0,036)
Homeowner	0,426 (0,495)	0,432 (0,497)	-0,006 (0,049)
House with water and sanitation connection	0,931 (0,253)	0,932 (0,252)	-0,001 (0,025)
House with electricity connection	0,946 (0,227)	0,963 (0,189)	-0,017 (0,021)
Father with no schooling	0,052 (0,222)	0,029 (0,168)	0,023 (0,022)
Father with primary schooling	0,231 (0,422)	0,261 (0,441)	-0,030 (0,045)
Father with secondary schooling	0,538 (0,500)	0,580 (0,495)	-0,042 (0,053)
Father with higher education	0,179 (0,384)	0,130 (0,338)	0,049 (0,039)

VARIABLES	(1) Admitted	(2) Not admitted	(3) Admitted versus not admitted ^a
Immigrant father	0,490 (0,501)	0,478 (0,501)	0,012 (0,053)
Mother with no schooling	0,059 (0,236)	0,064 (0,246)	-0,005 (0,024)
Mother with primary schooling	0,232 (0,423)	0,353 (0,479)	-0,120** (0,045)
Mother with secondary schooling	0,557 (0,498)	0,519 (0,501)	0,038 (0,050)
Mother with higher education	0,151 (0,359)	0,064 (0,246)	0,087** (0,032)
Immigrant mother	0,458 (0,499)	0,455 (0,500)	0,002 (0,050)
Number of observations	277	162	439

Note: Robust standard errors in parenthesis (*** p<0.01, ** p<0.05, *p<0.1).

^a Column (3) shows the regression coefficient for the variable shown in each row over a constant and a dummy variable equal to one if the student won the lottery and zero otherwise.

In addition, we estimated the probability of winning the lottery compared to all other characteristic variables. No significant coefficient was found at 95% confidence except for the presence of the father in the household (a positive correlation with winning the lottery). Pseudo R2 was 0.042 and the likelihood of the Wald F test was 0.12.

Tables 2 and 3 show a set of descriptive statistics on reading comprehension and math performance. These statistics were gathered after the lottery for winning and non-winning students in years 2007, 2008, 2009, 2010 and 2012. Column 3 shows the differences in the means for winning and for non-winning students. The means show that winning students have performance higher than non-winning ones, and this difference has grown over time.

Two equations were estimated to determine the impact of attending the FyA school on math performance and reading comprehension. The first equation is as follows:

$$y_i = \beta_0 + \rho F_i + \gamma_t + u_i \quad (1)$$

in which y_i is the performance of individual i that is standardized by subtracting the mean and by dividing the standard deviation by year, β_0 is a constant, γ_t is a fixed effect per year, F_i variable records whether the student was elected randomly to study at the FyA school, and u_{it} is an error term to capture random fluctuations in performance tests. The (average) causal effect is ρ .

Table 2: Descriptive statistics for reading comprehension performance

Year	Admitted	Not admitted	Admitted versus not admitted
2007	562,31 ^a	561,83	13,06 ^b
	(75,715)	(99,346)	(21,24)
	62	29	
2008	547,94	546,48	1,259
	(51,703)	(75,618)	(14,66)
	62	35	
2009	562,65	542,48	16,60
	(59,969)	(64,526)	(16,61)
	75	21	
2010	589,1	553,11	36,26**
	(67,739)	(65,813)	(14,01)
	60	38	
2012	655,07	563,55	91,52***
	(65,437)	(80,841)	(19,50)
	69	20	

Note: (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$

^a The order of the statistics are the mean, standard deviation, and the number of observations.

^b Differences are just for the mean.

Table 3: Descriptive statistics for math performance

Year	Admitted	Not admitted	Admitted versus not admitted
2007	539,58 ^a	527,46	29,83* ^b
	(80,826)	(89,312)	(16,91)
	62	28	
2008	506,19	552,86	-47,28***
	(49,548)	(71,350)	(13,87)
	62	35	
2009	595,37	550,67	39,85
	(97,421)	(111,736)	(28,65)
	75	21	
2010	581,92	518,92	63,73***
	(107,016)	(90,696)	(20,45)
	60	38	
2012	686,32	527,05	159,3***
	(94,475)	(87,582)	(22,43)
	69	20	

Note: (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$

^a The order of the statistics are the mean, standard deviation, and the number of observations.

^b Differences are just for the mean.

The second equation is as follows:

$$y_i = \beta_0 + \rho_{pub}Pub_i + \rho_{pri}Pri_i + \gamma_t + \varepsilon_i \quad (2)$$

in which Pub_i is a dichotomous variable that equals one if individual i loses the lottery in year t and attends another public school and zero otherwise, Pri_i is a dichotomous variable that equals one if the individual i loses the lottery in year t and attends a private school and zero otherwise, and ε_i is an error term to capture random fluctuations in the performance tests. Parameters ρ_{pub} and ρ_{pri} measure the effect of not attending a FyA school compared to attending a public school or a private school, respectively. Our first hypothesis of interest states that attending the charter school has a positive effect on the test scores for math and reading comprehension. That is, ρ , the causal effect of attending the charter school is positive ($\rho > 0$). Our second hypothesis is that the effect of FyA is higher when compared to private schools. That is ρ_{pub} and ρ_{pri} are both negative, and $|\rho_{pri}| > \rho$.

5 Results

We begin this section by testing our two main hypothesis. First, we present our main results that respond to the question of whether attending the FyA school has a positive effect on school performance. That is, the results of estimating equation (1). Then, we present evidence that the FyA students' performance is above the performance of comparable private school students. These results correspond to the estimation of equation (2).

5.1 Main results

Table 4 shows that the winning students reached performance levels of 0.39 and 0.41 standard deviations above those of non-winning students in reading comprehension and math respectively. This is a remarkable result because they are both above the findings by Angrist et al. (2010) for KIPP schools in Boston and by Abdulkadiroglu, Angrist, Dynarski, Kane, & Pathak (2009) for Boston's schools.

5.2 Heterogeneity according to public or private schools

We also compared the results obtained for FyA to the outcomes in public and private schools. Although it is true that for non-winning students attending a public or private school is not a lottery-driven decision, these results can explain the origin of the main differences between them. For their reading comprehension and math winning

students were 0.40 and 0.36 s.d., respectively, above non-winning students who attended a public school. For their reading comprehension and math, winning students performed 0.37 and 0.5 s.d., respectively, above non-winning students who later attended a private school. These results show that the private schools attended by some of the non-winning students were worse than the other public schools the remaining non-winning students attended, particularly in math. These results contradict the parents' perception that private schools provide a higher quality education. Since no lottery helps to determine which of the non-winners attend a public or private school. If we argue that those students who attend a private school come from families who have marginally higher incomes, then we can say that their children are marginally more skilled and therefore, the identified effect could be mitigated by these greater skills. Consequently, the estimate sets a lower bound for the negative effect of attending a private school but even a larger negative effect when controlling for skills. A reverse finding would result from a comparison with other public schools: a lower difference between FyA and other public schools should exist when controlling for skills.

Table 4: Performance estimates

VARIABLES	Reading comprehension		Math	
	(1)	(2)	(1)	(2)
FyA school	0,386*** (0,103)		0,413*** (0,0913)	
Private school		-0,371** (0,158)		-0,504*** (0,116)
Public school		-0,395*** (0,133)		-0,360** (0,136)
Observations	462	462	462	462
R-square	0,191	0,191	0,225	0,227

Note: Robust standard errors are in parenthesis (*** p<0.01, ** p<0.05, * p<0.1).

Regressions include year fixed effects and the constant.

5.3 Heterogeneity by quartile of performance

We estimated quartiles to determine whether there are heterogeneous effects by level of performance, as shown in Tables 5 and 6. The lowest and highest quartile showed that students had lower performance than FyA students (0.52 and 0.49 for the lowest quartile in reading comprehension and math, respectively, compared to 0.17 and 0.24 for the highest quartile for reading comprehension and math, respectively). This

impact may be due to FyA's efforts to level off classes by providing additional support to lagging students.

The results do not change when we compare the types of schools attended by non-winners. FyA performs better than public and private schools for all quartiles. In particular, students in the lowest quartile did much better at FyA than those in the highest quartile; students who attended a public school perform worse than students who attended a private school in reading comprehension for all quartiles.

Table 5: Reading comprehension performance (Quantile regression)

VARIABLES	(1) Lowest Quartile	(2) Highest Quartile	(3) Lowest Quartile	(4) Highest Quartile
FyA 26 school	0,520*** (0,151)	0,169*** (0,0282)		
Private school			-0,416** (0,192)	-0,169** (0,0543)
Public school			-0,663*** (0,160)	-0,182*** (0,0435)
Observations	462	462	462	462
Pseudo R-square	0,0772	0,1460	0,0790	0,1460

Robust standard errors are in parenthesis (*** p<0.01, ** p<0.05, * p<0.1).

Regressions include year fixed effects and the constant.

Table 6: Mathematics performance (Quantile regression)

VARIABLES	(1) Lowest Quartile	(2) Highest Quartile	(3) Lowest Quartile	(4) Highest Quartile
FyA 26 school	0,488*** (0,106)	0,239* (0,135)		
Private school			-0,478*** (0,165)	-0,373*** (0,105)
Public school			-0,593*** (0,136)	-0,134* (0,0790)
Observations	462	462	462	462
Pseudo R-square	0,091	0,1626	0,0913	0,1647

Robust standard errors are in parenthesis (*** p<0.01, ** p<0.05, * p<0.1).

Regressions include year fixed effects and the constant.

However, in math, students who attended a public school perform worse than students who attended a private school just for the lowest quartile. We observe the inverse in the highest quartile.

5.4 Heterogeneity over years of education

We estimate the effects by year to determine whether the impact of attending the FyA school changed over time. Our findings (Table 7) show on average the FyA school under study had a positive and significant effect both on reading comprehension and math with total 0.17 and 0.29 s.d., respectively, for 2007. The marginal effects are positive and increase over time (except for the negative outcome for math in 2008), although they are not statistically significant. Further, in 2012 the total effect reached 1.02 and 1.2 s.d. for reading comprehension and math. This result may be due to the fact that the census evaluation of the second grade students used in this study was administered for the first time in 2006, but it has become increasingly relevant over the years as it gained more public attention (especially given the attention given to it by the media). Possibly the interest in this Census evaluation started in this school in 2009 and plans were developed with teachers to engage in pedagogical activities aimed at enhancing the student body's performance. These findings are surprising because they are 10 and 5 times larger than the effects identified by Angrist et. al. (2010) for reading comprehension and math, respectively.

Table 7: Estimates for interactions

VARIABLES	Reading comprehension	Math
	(1)	(1)
FyA school	0,170 (0,241)	0,285** (0,137)
FyA*2008	-0,153 (0,265)	-0,738*** (0,208)
FyA*2009	0,0461 (0,284)	0,0958 (0,281)
FyA*2010	0,302 (0,290)	0,324 (0,200)
FyA*2012	1,021*** (0,266)	1,238*** (0,224)
Observations	462	462
R-square	0,222	0,305

Robust standard errors are in parenthesis (*** p<0.01, ** p<0.05, * p<0.1).

Regressions include the fixed effect and the constant.

The year 2011 is not in the regression because that year there was no test.

5.5 Main mechanisms

There are two main mechanisms through which FyA generates such an important effect. The first one is the high ability of directors chosen by FyA, in particular with better pedagogical and management skills. Better management results in better equipment (computer room, library), better infrastructure (students with tables, classrooms without requiring repairs), and more public services (electricity, water and sanitation). The effect of these components can be seen in table 8. The second mechanism was an important change in 2010. Based on 2007-2009 tests, FyA changed the deputy director and devoted more effort to preparing students for the test. In particular, they started with mock tests at least three times a year before the regular test in November of each year.

Table 8: Descriptive statistics of the variables in the Educational Census

VARIABLE	(1) Admitted	(2) Not admitted	(3) Admitted versus not admitted ^a
Director with pedagogical studies	Yes	0,918 (0,276)	0,0919*** (0,024)
Computer Room at school	Yes	0,776 (0,418)	0,223*** (0,036)
School library	Yes	0,649 (0,479)	0,351*** (0,040)
Students without folder / chair / table at school	No	1,731 (5,368)	-1,803*** (0,472)
Classrooms requiring repairs	2,076 (1,664)	7,276 (12,95)	-5,429*** (1,075)
Children in school	Yes	0,821 (0,385)	0,176*** (0,032)
Electricity in school	Yes	0,813 (0,391)	0,183*** (0,032)
Drinking water at school	Yes	0,821 (0,385)	0,175*** (0,031)
Sanitation at school	Yes		

Note: Robust standard errors are in parenthesis (*** p<0.01, ** p<0.05, *p<0.1).

^a Column (3) shows the regression coefficient for the variable shown in each row over a constant and a dummy variable equal to one if the student won the lottery and zero otherwise.

In addition, we estimated the probability of winning the lottery compared to all other characteristic variables. No significant coefficient was found at 95% confidence except for the presence of the father in the household (a positive correlation with winning the lottery). Pseudo R² was 0.042 and the likelihood of the Wald F test was 0.12.

6 Conclusions

The role of public-private partnerships in education may increase quality and may reduce inequality in developing countries. This is a matter of great policy relevance. In this study we have estimated the effect of a large worldwide public-private partnership, *Fe y Alegría*, on the achievement of relatively poor students in second grade. Our estimates show that the FyA school under analysis accomplished a significant gain in learning and performance in both math and reading comprehension (0.4 standard deviations). Although we could not determine the exact initial characteristics of winning students compared to non-winning ones due to insufficient information, the characteristics of the schools that both groups attended were used as a proxy for this analysis.

The main differences we find between the FyA school and others is basically the director. The study finds no significant differences between the FyA school in this study and the schools attended by the non-winning students (except for facilities including libraries and computer rooms). In addition, our analysis by quartiles determined that the positive impact of FyA school on learning is higher in the lower performance quartiles (approximately 0.5 standard deviations) and has a larger difference in reading comprehension (0.52 standard deviations). Another interesting finding is that the FyA school in this study has a larger impact on math when compared to a private school (0.5 standard deviations). Finally, the positive effect of the FyA has increased over time, particularly in 2012. Reading comprehension rose from 0.17 s.d. in 2007 to 1.02 s.d. in 2012. In math, its impact rose from 0.29 in 2007 to 1.2 s.d. in 2012. Estimating this effect of the FyA school under study is significant because it shows a much larger gain than so far reported in the literature (Angrist et al. 2010). No international study has found as large an effect as the one estimated in this study.

Our findings point to the need for more research on how classroom management evolved in those years, and for data from lottery outcomes at other FyA schools with a view at identifying more general effects. Consequently, on the basis of the evidence shown in this and previous studies, we suggest an in-depth examination of the possibility to adapt the pedagogical and administrative management models enforced in FyA schools and adopt their good practices in other public and private schools nationwide, especially those serving relatively poor, urban populations such as the one studied here.

The main limitation of this study is that it focuses on just a single FyA school, although it reflects an average FyA institution. Expanding this research to other FyA schools in Peru and other countries would seem an important topic for future research.

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Appendix: Distribution of students excluded from our analysis

In 2007, 2008, 2009, 2010, and 2012, we only include 462 students in our analysis out of a total 543 who joined the lottery to gain admittance in the FyA. Of all winning and non-winning students, there are 72 missing values in the database. Their distribution by year and sex is as follows:

Table A1: Total students without Student Census Evaluation data

Year	Female		Male		Total
	Winner	Non-winner	Winner	Non-winner	
2006	8	3	2	6	19
2007	4	4	5	6	19
2008	2	5	3	3	13
2009	4	5	1	2	12
2011	4	1	2	2	9
Total	22	18	13	19	72

Likewise, since six out of the total non-winning students migrated to other schools outside the San Juan de Lurigancho jurisdiction, we excluded these observations from our analysis, because those students' characteristics are presumably different from those of the other non-winning students who enrolled at some other school in San Juan de Lurigancho. Likewise, two students in the non-winners group chose to migrate to other FyA schools in San Juan de Lurigancho, where they gained admission. To isolate the impact of FyA, we chose to remove these observations. Finally, we also removed data for one non-winning students for whom there is no record of grades in mathematics. Our initial database screening yielded a total 462 students, whose data were used for the statistical and econometric analyses.